

CHAPTER 10

NUMERICAL DIFFERENTIATION

10.1 INTRODUCTION

In the solution of many practical problems, we are often concerned to find the derivatives of desired orders of a function of the form $y = f(x)$ given in tabular form without knowing the functional relationship between these two variables. The analytical process of differentiation fails in this case. Polynomial approximation methods described in chapters 4 and 5 provide a foundation for numerical differentiation in this regard. We assume that the data can be represented by a polynomial in the region of interest. We could then construct the polynomial, using one of the interpolation formulae discussed in chapters 4 and 5, and differentiate it (analytically) as many times as desired.

If the data is equally spaced, we use interpolating formula employing differences, such as Newton's, Stirling's, or Bessel's. If we desire the derivative at a point near the beginning of a tabulated data we use Newton's forward difference formula. However, if we desire the derivative at a point near the end of the table, we use Newton's backward difference formula. For points near the middle of the table we use a central difference formula such as Stirling's or Bessel's.

If the data is not equally spaced, we can use Lagrange's formula (section 4.2) or Newton's formula (section 4.4.2).

Suppose that the given function $f(x)$ is replaced by the interpolation polynomial $P(x)$ on the interval $[a, b]$, the region of our interest. Then

$$f'(x) \approx P'(x), \quad \text{for } a \leq x \leq b.$$

We can find the higher-order derivatives of $f(x)$ in the same way. If

$$R(x) = f(x) - P(x),$$

is the error in the interpolation polynomial $P(x)$, then the error of the derivative $P'(x)$ is given by the relation

$$r(x) = R'(x) = f'(x) - P'(x).$$

Thus the error of the derivative of an interpolation polynomial is equal to the derivative of the error in that interpolation polynomial. This is also true for the higher-order derivatives.

10.2 NUMERICAL DIFFERENTIATION FORMULAE BASED ON EQUALLY SPACED DATA

10.2.1 NUMERICAL DIFFERENTIATION BASED ON NEWTON'S FORWARD DIFFERENCES

Consider a function $y(x)$ specified at equally spaced $(n+1)$ points $x_i (i=0,1,2,\dots,n)$ in some interval $[a,b]$ by the relation

$$y_i = f(x_i).$$

We assume that the appropriate derivatives of $f(x)$ exist. Replacing the function $y(x)$ by Newton's interpolation polynomial (see section 5.2) we have

$$y_p \approx y_0 + p\Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!} \Delta^4 y_0 + \dots,$$

or

$$y_p \approx y_0 + p\Delta y_0 + \frac{p^2-p}{2!} \Delta^2 y_0 + \frac{p^3-3p^2+2p}{3!} \Delta^3 y_0 + \frac{p^4-6p^3+11p^2-6p}{4!} \Delta^4 y_0 + \dots, \quad (10.1)$$

where

$$p = \frac{x-x_0}{h} \text{ and } h = x_{i+1} - x_i \quad (i = 0,1,2,\dots).$$

$$\text{Now } \frac{dy}{dx} = \frac{dy}{dp} \times \frac{dp}{dx} = \frac{dy}{dp} \left(\frac{1}{h}\right) = \frac{1}{h} \frac{dy}{dp}.$$

From (10.1) we get

$$\frac{dy}{dx}_p = y'_p \approx \frac{1}{h} \frac{d}{dp} \left[y_0 + p\Delta y_0 + \frac{p^2-p}{2!} \Delta^2 y_0 + \frac{p^3-3p^2+2p}{3!} \Delta^3 y_0 + \dots \right]$$

$$y'_p \approx \frac{1}{h} \left[\Delta y_0 + \frac{2p-1}{2!} \Delta^2 y_0 + \frac{3p^2-6p+2}{3!} \Delta^3 y_0 + \dots \right] \quad (10.2)$$

Since

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left[\frac{1}{h} \frac{dy}{dp} \right] = \frac{dp}{dx} \frac{d}{dp} \left[\frac{1}{h} \frac{dy}{dp} \right] = \frac{1}{h^2} \frac{d^2 y}{dp^2},$$

In general
$$\frac{d^n y}{dx^n} = \frac{1}{h^n} \frac{d^n y}{dp^n}.$$

Thus
$$y''_p \approx \frac{1}{h^2} \left[\Delta^2 y_0 + (p-1) \Delta^3 y_0 + \frac{6p^2-18p+11}{12} \Delta^4 y_0 + \dots \right] \quad (10.3)$$

$$y'''_p \approx \frac{1}{h^3} \left[\Delta^3 y_0 + \frac{2p-3}{2} \Delta^4 y_0 + \dots \right] \quad (10.4)$$

$$y^{IV}_p \approx \frac{1}{h^4} [\Delta^4 y_0 + \dots] \quad (10.5)$$

The formulae (10.2)-(10.5) are used to find derivatives at an interpolated point i.e. at a point other than a tabulated value of x . When seeking the derivatives $y'_p, y''_p, \dots$ at fixed point x , one should choose for x_0 the closest tabular value of the argument.

NUMERICAL DIFFERENTIATION AT TABULATED POINTS:

It is often required to find the derivatives at tabulated value of x . The numerical differentiation formulae are simplified in this case. Each tabular value may be taken as the initial value, we set $x = x_0$, then $p = 0$. Thus

$$(y')_{x_0} \approx \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right] \quad (10.2')$$

$$(y'')_{x_0} \approx \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \dots \right] \quad (10.3')$$

$$(y''')_{x_0} \approx \frac{1}{h^3} \left[\Delta^3 y_0 - \frac{3}{2} \Delta^4 y_0 + \dots \right] \quad (10.4')$$

DERIVATIVES IN TERMS OF FUNCTIONAL VALUES:

The numerical differentiation formula can be expressed in terms of functional values instead of differences. We can obtain the numerical differentiation formulae involving any desired number of data points, by truncating (10.2')-(10.4') after the appropriate term.

i) Two-point forward formula.

Retaining only the first term of (10.2') we get

$$y'_{x_0} \approx \frac{1}{h} [\Delta y_0] = \frac{1}{h} [y_1 - y_0] \quad (10.6)$$

ii) Three-point forward formula.

Retaining the first two terms of (10.2') we obtain

$$\begin{aligned} y'_{x_0} &\approx \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 \right] \\ &= \frac{1}{h} \left[(y_1 - y_0) - \frac{1}{2} \Delta(y_1 - y_0) \right] \\ &= \frac{1}{h} \left[y_1 - y_0 - \frac{1}{2} ((y_2 - y_1) - (y_1 - y_0)) \right] \\ &= \frac{1}{2h} [-y_2 + 4y_1 - 3y_0] \end{aligned} \quad (10.7)$$

ii) Four-point forward formula.

$$\begin{aligned} y'_{x_0} &\approx \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 \right] \\ &= \frac{1}{h} \left[y_1 - y_0 - \frac{1}{2} (y_2 - 2y_1 + y_0) + \frac{1}{3} ((y_3 - y_2) - 2(y_2 - y_1) + (y_1 - y_0)) \right] \\ &= \frac{1}{h} \left[y_1 - y_0 - \frac{1}{2} (y_2 - 2y_1 + y_0) + \frac{1}{3} (y_3 - 3y_2 + 3y_1 - y_0) \right] \\ &= \frac{1}{6h} [6y_1 - 6y_0 - 3y_2 + 6y_1 - 3y_0 + 2y_3 - 6y_2 + 6y_1 - 2y_0] \\ &= \frac{1}{6h} [2y_3 - 9y_2 + 18y_1 - 11y_0] \end{aligned} \quad (10.8)$$

EXAMPLE 10.1: Find the first and second derivatives of the function tabulated below at the points $x = 2.31$ and 3.0 .

x	0	1	2	3	4	5	6
y	2	3	10	29	66	127	218

SOLUTION: Construct a difference table as given below.

Table 10.1 Difference Table for Derivative Approximation

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0	2				
1	3	1			
2	10	7	6		
3	29	19	12	6	
4	66	37	18	6	0
5	127	61	24	6	0
6	218	91	30	6	0

When $x = 2.31$.

Take $x_0 = 2$, then $p = 0.31$.

Here $h = 1$, $y_0 = 10$, $\Delta y_0 = 19$, $\Delta^2 y_0 = 18$, $\Delta^3 y_0 = 6$, $\Delta^4 y_0 = 0$.

Using formulae (10.2) and (10.3) we obtain

$$\begin{aligned}
 y'_p &\approx \frac{1}{h} \left[19 + \frac{2(0.31) - 1}{2} 18 + \frac{3(0.31)^2 - 6(0.31) + 2}{6} 6 \right] \\
 &= [19 - 3.42 + 0.4283] \\
 &= 16.0083 \approx 16.008
 \end{aligned}$$

and $y''_p \approx (1/h^2)[18 + (0.31 - 1)6] = 13.86$.

When $x = 3$.

In this case we choose $x_0 = 3$, then

$$\Delta y_0 = 37, \quad \Delta^2 y_0 = 24, \quad \Delta^3 y_0 = 6.$$

Using formulae (10.2') and (10.3') we get

$$y'_{x_0} \approx \frac{1}{1} \left[37 - \frac{1}{2} (24) + \frac{1}{3} (6) \right]$$

$$= 37 - 12 + 2 = 27$$

and

$$y''_{x_0} \approx \frac{1}{1^2} [24 - 6] = 18.$$

To estimate the accuracy of these values, we note that as the table represents the polynomial $y = x^3 + 2$,

$$y' = 3x^2, \quad y'' = 6x.$$

Hence

$$y'(2.31) = 3(2.31)^2 = 16.0083 \approx 16.008$$

$$y'(3) = 3(3)^2 = 27$$

$$y''(2.31) = 6(2.31) = 13.86$$

$$y''(3) = 6(3) = 18.$$

Thus the results are correct to the given digits.

10.2.2 NUMERICAL DIFFERENTIATION BASED ON NEWTON'S BACKWARD DIFFERENCES

We know that Newton's backward difference formula (5.2) is

$$y_p \approx y_0 + p \nabla y_0 + \frac{p(p+1)}{2!} \nabla^2 y_0 + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_0 + \frac{p(p+1)(p+2)(p+3)}{4!} \nabla^4 y_0 + \dots$$

$$= y_0 + p \nabla y_0 + \frac{p^2 + p}{2!} \nabla^2 y_0 + \frac{p^3 + 3p^2 + 2p}{3!} \nabla^3 y_0 + \frac{p^4 + 6p^3 + 11p^2 + 6p}{4!} \nabla^4 y_0 + \dots$$

Therefore

$$y'_p \approx \frac{1}{h} \left[\nabla y_0 + \frac{1}{2} (2p+1) \nabla^2 y_0 + \frac{3p^2 + 6p + 2}{6} \nabla^3 y_0 + \frac{2p^3 + 9p^2 + 11p + 3}{12} \nabla^4 y_0 + \dots \right]$$

(10.9)

$$y''_p \approx \frac{1}{h^2} \left[\nabla^2 y_0 + (p+1) \nabla^3 y_0 + \frac{6p^2 + 18p + 11}{12} \nabla^4 y_0 + \dots \right]$$

(10.10)

$$y_p''' \approx \frac{1}{h^3} \left[\nabla^3 y_0 + \frac{1}{2} (2p + 3) \nabla^4 y_0 + \dots \right] \quad (10.11)$$

$$y_p^{iv} \approx \frac{1}{h^4} \left[\nabla^4 y_0 + \dots \right] \quad (10.12)$$

and so on.

If $p = 0$, we obtain

$$y_{x_0}' \approx \frac{1}{h} \left[\nabla y_0 + \frac{1}{2} \nabla^2 y_0 + \frac{1}{3} \nabla^3 y_0 + \frac{1}{4} \nabla^4 y_0 + \dots \right] \quad (10.9')$$

$$y_{x_0}'' \approx \frac{1}{h^2} \left[\nabla^2 y_0 + \nabla^3 y_0 + \frac{11}{12} \nabla^4 y_0 + \dots \right] \quad (10.10')$$

$$y_{x_0}''' \approx \frac{1}{h^3} \left[\nabla^3 y_0 + \frac{3}{2} \nabla^4 y_0 + \dots \right] \quad (10.11')$$

To express the derivatives in terms of functional values we retain one, two and three terms of (10.9') and obtain, respectively, two-point, three-point, and four-point backward formulae.

$$y_{x_0}' \approx \frac{1}{h} [y_0 - y_{-1}] \quad (10.13)$$

$$y_{x_0}' \approx \frac{1}{2h} [3y_0 - 4y_{-1} + y_{-2}] \quad (10.14)$$

$$y_{x_0}' \approx \frac{1}{6h} [11y_0 - 18y_{-1} + 9y_{-2} - 2y_{-3}] \quad (10.15)$$

EXAMPLE 10.2: Find the first and second derivatives of the function tabulated in Example 10.1 at the points $x = 5$ and 5.7 .

SOLUTION: To estimate the desired derivatives, we use the numerical differentiation formula based on Newton's backward differences.

When $x = 5$, we take $x_0 = 5$. Then

$$\nabla y_0 = 61, \nabla^2 y_0 = 24, \nabla^3 y_0 = 6, \nabla^4 y_0 = 0.$$

Thus the differentiation formulae (10.9') and (10.10') yield respectively

$$y'_{x_0} \approx \frac{1}{1} \left[61 + \frac{1}{2} (24) + \frac{1}{3} (6) \right] = 61 + 12 + 2 = 75$$

and

$$y''_{x_0} \approx \frac{1}{1} [24 + 6] = 30.$$

When $x = 5.7$.

Take $x_0 = 6$, then $p = 5.7 - 6 = -0.3$, and

$$\nabla y_0 = 91, \nabla^2 y_0 = 30, \nabla^3 y_0 = 6, \nabla^4 y_0 = 0.$$

Using (10.9) and (10.10) respectively, we obtain

$$\begin{aligned} y'_p &\approx \frac{1}{1} \left[91 + \frac{1}{2} (-0.6 + 1) 30 + \frac{3(0.09) - 1.8 + 2}{6} 6 \right] \\ &= [91 + 6 + 0.47] = 97.47 \end{aligned}$$

and

$$y''_p \approx \frac{1}{1^2} [30 + (-0.3 + 1) 6] = 30 + 4.2 = 34.2.$$

Let us estimate the accuracy of these results. We know that

$$y = x^3 + 2.$$

Thus

$$y'_x = 3x^2, y''_x = 6x.$$

Therefore

$$\begin{aligned} y'(5) &= 3(25) = 75 \\ y'(5.7) &= 3(32.49) = 97.47 \\ y''(5) &= 6(5) = 30 \\ y''(5.7) &= 6(5.7) = 34.2. \end{aligned}$$

Hence the results are accurate to the given digits.

10.2.3 NUMERICAL DIFFERENTIATION BASED ON STIRLING'S FORMULA

The numerical differentiation formulae discussed in Sec. (10.2.1) & (10.2.2) for a function $y(x)$ at a point $x = x_0$ have the disadvantage that they only take one-sided values of the function. The formulae of sec. (10.2.1) employ values of the function for $x > x_0$ while the formulae of sec. (10.2.2) employ values of the function for $x < x_0$. However, a relatively higher accuracy is

ensured by the symmetric formulae of differentiation which employ the values of the given function $y(x)$ both for $x > x_0$ and $x < x_0$. These formulae are generally called central difference formulae. Any one of the central difference interpolation formulae can be used to give a formula for derivatives. We will derive only the formulae based on Stirling's and Bessel's interpolation formulae.

Let $\dots, x_{-2}, x_{-1}, x_0, x_1, x_2, \dots$ be a set of equally spaced points with $x_{i+1} = x_i + h$. Suppose that $y_i = f(x_i)$ give the corresponding values of the given function $y=f(x)$. Replacing the given function y by the Stirling's interpolation polynomial (Sec.5.5) and setting $p = (x - x_0)/h$, we obtain

$$y_p \approx y_0 + p\mu\delta y_0 + \frac{p^2}{2!} \delta^2 y_0 + \frac{p(p^2-1)}{3!} \mu\delta^3 y_0 + \frac{p^2(p^2-1)}{4!} \delta^4 y_0 + \dots$$

or

$$y_p \approx y_0 + p \frac{\Delta y_{-1} + \Delta y_0}{2} + \frac{p^2}{2!} \Delta^2 y_{-1} + \frac{p(p^2-1)}{3!} \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} +$$

$$\frac{p^2(p^2-1)}{4!} \Delta^4 y_{-2} + \frac{p(p^2-1)(p^2-2^2)}{5!} \frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \dots$$

Since $\frac{dy}{dx} = \frac{dy}{dp} \frac{dp}{dx} = \frac{1}{h} \frac{dy}{dp}$, thus from the above relation we get

$$y'_p \approx \frac{1}{h} \left[\frac{\Delta y_{-1} + \Delta y_0}{2} + p \Delta^2 y_{-1} + \frac{3p^2-1}{3!} \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \frac{4p^3-2p}{4!} \Delta^4 y_{-2} \right. \\ \left. + \frac{5p^4-15p^2+4}{5!} \frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \dots \right] \quad (10.16)$$

$$y''_p \approx \frac{1}{h^2} \left[\Delta^2 y_{-1} + p \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \frac{6p^2-1}{12} \Delta^4 y_{-2} + \frac{2p^3-3p}{12} \frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} \right. \\ \left. + \dots \right] \quad (10.17)$$

$$y'''_p \approx \frac{1}{h^3} \left[\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + p \Delta^4 y_{-2} + \frac{2p^2-1}{4} \frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \dots \right] \quad (10.18)$$

$$y_p^{IV} \approx \frac{1}{h^4} \left[\Delta^4 y_{-2} + p \frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \dots \right] \quad (10.19)$$

$$y_p^V \approx \frac{1}{h^5} \left[\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \dots \right]. \quad (10.20)$$

If we take $x=x_0$, then $p=0$ and formulae (10.16) - (10.20) take the form

$$y'_{x_0} \approx \frac{1}{h} \left[\frac{\Delta y_{-1} + \Delta y_0}{2} - \frac{1}{3!} \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \frac{4}{5!} \frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \dots \right] \quad (10.16')$$

$$y''_{x_0} \approx \frac{1}{h^2} \left[\Delta^2 y_{-1} - \frac{1}{12} \Delta^4 y_{-2} + \dots \right] \quad (10.17')$$

$$y'''_{x_0} \approx \frac{1}{h^3} \left[\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} - \frac{1}{4} \frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \dots \right] \quad (10.18')$$

$$y_{x_0}^{IV} \approx \frac{1}{h^4} \left[\Delta^4 y_{-2} + \dots \right] \quad (10.19')$$

$$y_{x_0}^V \approx \frac{1}{h^5} \left[\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \dots \right] \quad (10.20')$$

EXAMPLE 10.3: Find $y'(0.6)$ and $y''(0.6)$ for the function $y = y(x)$ given by the following table.

x	0.4	0.5	0.6	0.7	0.8
y	1.5836494	1.7974426	2.0442376	2.3275054	2.6510818

SOLUTION: First of all we construct the difference Table 10.2.

Take $x_0 = 0.6$, then $p = 0$, $h = 0.1$,

$$\Delta y_0 = 0.2832678, \Delta y_{-1} = 0.2467950, \Delta^2 y_{-1} = 0.0364728,$$

$$\Delta^3 y_{-1} = 0.0038358, \Delta^3 y_{-2} = 0.0034710, \Delta^4 y_{-2} = 0.0003648.$$

Table 10.2 Difference Table for Derivative Approximation

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.4	1.5836494				
		0.2137932			
0.5	1.7974426		0.0330018		
		0.2467950		0.0034710	
0.6	2.0442376		0.0364728		0.0003648
		0.2832678		0.0038358	
0.7	2.3275054		0.0403086		
		0.3235764			
0.8	2.6510818				

Substituting these values in the formulae (10.16') and (10.17'), respectively we get

$$y'_{x_0} \approx 10 \left[\frac{0.2467950 + 0.2832678}{2} - \frac{1}{6} \frac{0.0034710 + 0.0038358}{2} \right]$$

$$= 10[0.2650314 - 0.0006089] = 2.6442$$

$$y''_{x_0} \approx 100 \left[0.0364728 - \frac{1}{12} (0.0003648) \right]$$

$$= 100[0.0364728 - 0.0000304] = 3.6442.$$

As a check, we note that the tabulated function is

$$y = 2e^x - x - 1.$$

Thus $y' = 2e^x - 1$

and $y'' = 2e^x.$

Substituting $x = 0.6$, we obtain

$$y' = 2.6442, y'' = 3.6442.$$

By comparison of these values we note that the values found by numerical differentiation are correct to four decimal places in both cases.

10.2.4 NUMERICAL DIFFERENTIATION BASED ON BESSEL'S FORMULA

From Bessel's interpolation formula (5.6), we have

$$y_p \approx \frac{y_0 + y_1}{2} + (p-1/2)\Delta y_0 + \frac{p(p-1)}{2} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{p(p-1)(2p-1)}{12} \Delta^3 y_{-1} \\ + \frac{p(p-1)(p+1)(p-2)}{24} \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots$$

or

$$y_p \approx \frac{y_0 + y_1}{2} + (p-1/2)\Delta y_0 + \frac{p^2 - p}{2} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \left(\frac{1}{6} p^3 - \frac{1}{4} p^2 + \frac{1}{12} p \right) \Delta^3 y_{-1} + \\ \left(\frac{1}{24} p^4 - \frac{1}{12} p^3 - \frac{1}{24} p^2 + \frac{1}{12} p \right) \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots$$

Hence

$$y'_p \approx \frac{1}{h} \left[\Delta y_0 + \frac{2p-1}{2} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \left(\frac{1}{2} p^2 - \frac{1}{2} p + \frac{1}{12} \right) \Delta^3 y_{-1} + \right. \\ \left. \left(\frac{1}{6} p^3 - \frac{1}{4} p^2 - \frac{1}{12} p + \frac{1}{12} \right) \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \right] \quad (10.21)$$

$$y''_p \approx \frac{1}{h^2} \left[\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + (p - \frac{1}{2}) \Delta^3 y_{-1} + \left(\frac{1}{2} p^2 - \frac{1}{2} p - \frac{1}{12} \right) \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \right] \quad (10.22)$$

$$y'''_p \approx \frac{1}{h^3} \left[\Delta^3 y_{-1} + (p - \frac{1}{2}) \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \right] \quad (10.23)$$

$$y^{IV}_p \approx \frac{1}{h^4} \left[\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \right] \quad (10.24)$$

If $x = x_0$, then $p = 0$ and these formulae become

$$y'_{x_0} \approx \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} + \frac{1}{12} \Delta^3 y_{-1} + \frac{1}{12} \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \right] \quad (10.21')$$

$$y''_{x_0} \approx \frac{1}{h^2} \left[\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} - \frac{1}{2} \Delta^3 y_{-1} - \frac{1}{12} \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \right] \quad (10.22')$$

$$y'''_{x_0} \approx \frac{1}{h^3} \left[\Delta^3 y_{-1} - \frac{1}{2} \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \right] \quad (10.23')$$

$$y^{IV}_{x_0} \approx \frac{1}{h^4} \left[\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots \right] \quad (10.24')$$

EXAMPLE 10.4: Find $y'(0.6)$ and $y''(0.6)$ with the help of numerical differentiation formula based on Bessel's formula using the table of previous example.

SOLUTION: Here $x_0 = 0.6$, $p = 0$, $h = 0.1$. Thus

$$\begin{aligned} y'_{x_0} &\approx 10 \left[0.2832678 - \frac{1}{2} \frac{0.0364728 + 0.0403086}{2} + \frac{1}{12} 0.0038358 \right] \\ &= 10[0.2832678 - 0.0191954 + 0.0003196] \\ &= 2.643920. \end{aligned}$$

$$\begin{aligned} y''_{x_0} &\approx 100 \left[\frac{0.0364728 + 0.0403086}{2} - \frac{1}{2} (0.0038358) \right] \\ &= 100[0.0383907 - 0.0019179] \\ &= 3.64728. \end{aligned}$$

MAXIMA AND MINIMA:

It is sometimes required to find the maximum and minimum values of a differentiable function given in tabular form. For this purpose, we construct difference table from the given data and substitute them in the appropriate interpolation formula. Equate to zero the first derivative of this formula and solve for p . Then the value of x is found from the relation

$$x = x_0 + hp$$

10.3 NUMERICAL DIFFERENTIATION BASED ON LAGRANGE'S FORMULA

So far we have discussed various numerical differentiation formulae based on equally spaced data. It often happens that the data given in tabular form is unequally spaced and it is necessary, under certain circumstances, to use it for differentiation. Any of the interpolation formulae based on unequally spaced data can be used for this purpose. We introduce here only Lagrange's formulae.

Let $x_0, x_1, \dots, x_n$ be $(n+1)$ distinct points and $y=f(x)$ is a given function whose values are given at these points. We construct Lagrange's formula $P(x)$ for these set of points $x_i (i = 0, \dots, n)$ (Sec.4.2) such that

$$P(x) = L_0(x)y_0 + L_1(x)y_1 + \dots + L_n(x)y_n = \sum_{k=0}^n L_k(x)y_k \quad (10.25)$$

Where

$$L_k(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{k-1})(x-x_{k+1})\dots(x-x_n)}{(x_k-x_0)(x_k-x_1)\dots(x_k-x_{k-1})(x_k-x_{k+1})\dots(x_k-x_n)}$$

We note that

$$L_k(x_r) = \begin{cases} 1, & \text{if } r = k \\ 0, & \text{if } r \neq k \end{cases}$$

Differentiating (10.25) we obtain

$$P'(x) = \sum_{k=0}^n L'_k(x)y_k \quad (10.26)$$

$$P''(x) = \sum_{k=0}^n L''_k(x)y_k \quad (10.27)$$

$$P'''(x) = \sum_{k=0}^n L'''_k(x)y_k \quad (10.28)$$

To estimate the error, we use the formula (4.6)

$$R_n(x) = f(x) - p(x) = \frac{f^{(n+1)}(\xi)F(x)}{(n+1)!}$$

where

$$F(x) = (x - x_0)(x - x_1) \dots (x - x_n),$$

and

$$x_0 < \xi < x_n.$$

Assuming that $f(x) \in C^{n+2}$, then

$$r(x) = R'_n(x) = \frac{1}{(n+1)!} \left[f^{n+1}(\xi) \frac{d}{dx} F(x) + F(x) \frac{d}{dx} f^{n+1}(\xi) \right]$$

is difficult to analyze because the dependence of ξ on x is not known and $\frac{d}{dx}[f^{n+1}(\xi)]$ cannot be evaluated. However, if x is one of the interpolating points i.e. $x = x_k (k=0,1,2,\dots,n)$ and assuming that $\frac{d}{dx}[f^{n+1}(\xi)]$ is bounded, then

$$F(x_k) \frac{d}{dx} [f^{n+1}(\xi_k)] = 0$$

$$\begin{aligned} \text{and } \frac{d}{dx} F(x) &= \lim_{x \rightarrow x_k} \frac{(x-x_0)(x-x_1) \dots (x-x_k)(x-x_{k+1}) \dots (x-x_n)}{x-x_k} \\ &= (x_k-x_0)(x_k-x_1) \dots (x_k-x_{k-1})(x_k-x_{k+1}) \dots (x_k-x_n) \end{aligned}$$

$$r(x_k) = \frac{1}{(n+1)!} (x_k-x_0)(x_k-x_1) \dots (x_k-x_{k-1})(x_k-x_{k+1}) \dots (x_k-x_n) f^{n+1}(\xi_k) \quad (10.29)$$

In the same way we can find the error estimate for the higher order derivatives.

EXAMPLE 10.5: Find the first and second order derivatives of the function tabulated in Example 4.1 at $x = 1.2$.

SOLUTION: Here the interpolation polynomial is

$$P(x) \approx 2x^3 - 3x^2 + x - 2.$$

Thus

$$P'(x) \approx 6x^2 - 6x + 1,$$

$$P''(x) \approx 12x - 6.$$

Hence

$$P'(1.2) \approx 6(1.2)^2 - 6(1.2) + 1 = 6(1.44) - 7.2 + 1 = 2.44$$

$$P''(1.2) \approx 12(1.2) - 6 = 14.4 - 6 = 8.4.$$

10.4 ERROR ANALYSIS OF DIFFERENTIATION FORMULAE

Consider a differentiable function $f(x)$. Then by definition

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h}$$

or

$$f'(x_0) \approx \frac{f(x_0+h) - f(x_0)}{h}$$

$$f(x_0+h) \approx f(x_0) + hf'(x_0)$$

which is the linear Taylor's expansion. From this observation one may conclude that Taylor's expansion provides the basis for numerical differentiation formulae.

Assume that appropriate derivatives of $f(x)$ exist. Then by the Taylor's polynomial with remainder terms, we get

$$f(x_0 + h) = f(x_0) + hf'(x_0) + \frac{f''(\xi)}{2}h^2, \quad x_0 < \xi < x_0 + h.$$

$$\text{Thus } f'(x_0) = \frac{f(x_0+h) - f(x_0)}{h} - \frac{h}{2}f''(\xi). \quad (10.30)$$

If $f''(x)$ is bounded near x_0 , then we may simply say that the error of (10.30) is $O(h)$. We can raise the order of the error by considering the quadratic Taylor's polynomial with remainder terms

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2 + \frac{f'''(\xi_1)}{6}h^3, \quad x_0 < \xi_1 < x_0+h$$

$$f(x_0-h) = f(x_0) - f'(x_0)h + \frac{f''(x_0)}{2}h^2 - \frac{f'''(\xi_2)}{6}h^3, \quad x_0-h < \xi_2 < x_0$$

Subtraction of these equations yields

$$f(x_0+h) - f(x_0-h) = 2f'(x_0)h + \frac{1}{6}[f'''(\xi_1) + f'''(\xi_2)]h^3$$

or

$$f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h} - \frac{1}{12} [f'''(\xi_1) + f'''(\xi_2)]h^2$$

If f''' is continuous on $[x_0-h, x_0+h]$, then by the Intermediate Value Theorem (Appendix A) there exists $\xi \in (x_0-h, x_0+h)$ such that

$$f'''(\xi_1) + f'''(\xi_2) = 2f'''(\xi).$$

Thus

$$f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h} - \frac{1}{6} f'''(\xi)h^2. \quad (10.31)$$

This is known as three-point formula.

Let us now find an approximation for second derivatives.

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2 + \frac{f'''(x_0)}{3!}h^3 + \frac{f^{IV}(\xi_1)}{4!}h^4,$$

$x_0 < \xi_1 < x_0+h$

and

$$f(x_0-h) = f(x_0) - f'(x_0)h + \frac{f''(x_0)}{2}h^2 - \frac{f'''(x_0)}{3!}h^3 + \frac{f^{IV}(\xi_2)}{4!}h^4,$$

$x_0-h < \xi_2 < x_0$

Adding these equations and simplifying, we obtain

$$f''(x_0) = \frac{f(x_0+h) - 2f(x_0) + f(x_0-h)}{h^2} - \frac{h^2}{24} [f^{IV}(\xi_1) + f^{IV}(\xi_2)].$$

Assuming that f^{IV} is continuous on $[x_0-h, x_0+h]$, then by the Intermediate Value Theorem (Appendix A), there exists $\xi \in (x_0-h, x_0+h)$ such that

$$f^{IV}(\xi_1) + f^{IV}(\xi_2) = 2f^{IV}(\xi).$$

Then the above equation takes the form

$$f''(x_0) = \frac{1}{h^2} [f(x_0+h) - 2f(x_0) + f(x_0-h)] - \frac{h^2}{12} f^{IV}(\xi). \quad (10.32)$$

or
$$f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h} - \frac{1}{12} [f'''(\xi_1) + f'''(\xi_2)]h^2$$

If f''' is continuous on $[x_0-h, x_0+h]$, then by the Intermediate Value Theorem (Appendix A) there exists $\xi \in (x_0-h, x_0+h)$ such that

$$f'''(\xi_1) + f'''(\xi_2) = 2f'''(\xi).$$

Thus
$$f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h} - \frac{1}{6} f'''(\xi)h^2. \quad (10.31)$$

This is known as three-point formula.

Let us now find an approximation for second derivatives.

As

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{f''(x_0)}{2}h^2 + \frac{f'''(x_0)}{3!}h^3 + \frac{f^{IV}(\xi_1)}{4!}h^4,$$

$x_0 < \xi_1 < x_0+h$

and

$$f(x_0-h) = f(x_0) - f'(x_0)h + \frac{f''(x_0)}{2}h^2 - \frac{f'''(x_0)}{3!}h^3 + \frac{f^{IV}(\xi_2)}{4!}h^4,$$

$x_0-h < \xi_2 < x_0$

Adding these equations and simplifying, we obtain

$$f''(x_0) = \frac{f(x_0+h) - 2f(x_0) + f(x_0-h)}{h^2} - \frac{h^2}{24} [f^{IV}(\xi_1) + f^{IV}(\xi_2)].$$

Assuming that f^{IV} is continuous on $[x_0-h, x_0+h]$, then by the Intermediate Value Theorem (Appendix A), there exists $\xi \in (x_0-h, x_0+h)$ such that

$$f^{IV}(\xi_1) + f^{IV}(\xi_2) = 2f^{IV}(\xi).$$

Then the above equation takes the form

$$f''(x_0) = \frac{1}{h^2} [f(x_0+h) - 2f(x_0) + f(x_0-h)] - \frac{h^2}{12} f^{IV}(\xi). \quad (10.32)$$

We see that the accuracy of both (10.31) and (10.32) is of $O(h^2)$. Similar analysis can be made of other formulae discussed in this chapter.

EXAMPLE 10.6: Let $f(x) = e^x$. Compute approximation to $f'(0)$ and $f''(0)$, using (10.30), (10.31), (10.32) for $h=0.1$ with the following data. Also calculate actual error.

x	-0.1	0.0	1.1
f(x)	0.904837	1.000000	1.105172

SOLUTION: Exact value: $f'(0) = 1.000000$
Using (10.30), we get

$$f'(0) \approx \frac{f(1.1) - f(0)}{h} = \frac{1.105172 - 1.000000}{0.1} = 1.05172.$$

Actual error = $1.00000 - 1.05172 = -0.05172$.
Using (10.31)

$$f'(0) \approx \frac{f(1.1) - f(-0.1)}{2h} = \frac{1.105172 - 0.904837}{0.2} = 1.001675.$$

Actual error = $1.000000 - 1.001675 = -0.001675$.

Using (10.32)

$$\begin{aligned} f''(0) &\approx \frac{1}{h^2} [f(1.1) - 2f(0) + f(-0.1)] \\ &= (1/0.01) [1.105172 - 2(1.0) + 0.904837] = 1.0009. \end{aligned}$$

Exact value: $f''(0) = 1.0000$
Actual error = $1.0000 - 1.0009 = -0.0009$.

10.5 RICHARDSON EXTRAPOLATION

It is sometimes not possible to obtain an answer to desired accuracy by decreasing h . In this case the Richardson extrapolation technique can be used. This technique combines a sequence of estimates of the derivatives to obtain a further estimate which is more accurate.

Suppose that appropriate derivatives of $f(x)$ exist. Then by Taylor's series we can write

$$f(x_0 + h) = f(x_0) + hf'(x_0) + \frac{h^2}{2!}f''(x_0) + \frac{h^3}{3!}f'''(x_0) + \dots \quad (10.33)$$

and

$$f(x_0 - h) = f(x_0) - hf'(x_0) + \frac{h^2}{2!}f''(x_0) - \frac{h^3}{3!}f'''(x_0) + \dots \quad (10.34)$$

Subtracting (10.34) from (10.33) and dividing by $2h$ gives the relation

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0 - h)}{2h} - \frac{h^2}{3!}f'''(x_0) - \frac{h^4}{5!}f^{(5)}(x_0) + \dots \quad (10.35)$$

or

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0 - h)}{2h} - \sum_{i=1}^{\infty} \frac{h^{2i}}{(2i+1)!} f^{(2i+1)}(x_0). \quad (10.35')$$

Similarly we can obtain

$$f''(x_0) = \frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h))}{h^2} - \sum_{i=1}^{\infty} \frac{h^{2i}}{(2i+2)!} f^{(2i+2)}(x_0). \quad (10.36)$$

Replacing h in (10.35) by qh where q is any value other than ± 1 , then

$$f'(x_0) = \frac{f(x_0 + qh) - f(x_0 - qh)}{2qh} - \frac{q^2 h^2}{3!} f'''(x_0) - \frac{q^4 h^4}{5!} f^{(5)}(x_0) + \dots \quad (10.37)$$

Using (10.35) and (10.37) to eliminate the term involving h^2 to obtain

$$\begin{aligned} f'(x_0) &= \frac{1}{2hq(q^2 - 1)} \left[q^3 f(x_0 + h) - f(x_0 + qh) - q^3 f(x_0 - h) + f(x_0 - qh) \right] \\ &\quad + \frac{q^2 h^4}{5!} f^{(5)}(x_0) + \dots \end{aligned} \quad (10.38)$$

This gives an expression for $f'(x_0)$ that is accurate to order h^4 rather than h^2 . This process can be repeated by combining yet more function values to remove further terms from the error expansion. In this way successive approximations with error of

order h^6 , h^8 , etc., can be obtained.

Similarly we can obtain from (10.36)

$$f''(x_0) = \frac{1}{h^2 q^2 (q^2 - 1)} \left[q^4 f(x_0 + h) - f(x_0 + qh) - (2q^4 - 2)f(x_0) + q^4 f(x_0 - h) - f(x_0 - qh) \right] + \frac{q^2 h^4}{6!} f^{(6)}(x_0) + \dots \quad (10.39)$$

which is also accurate to order h^4 .

EXAMPLE 10.7: Find $y'(0.6)$ and $y''(0.6)$ for the function $y = y(x)$ using the data of the Example 10.3.

SOLUTION: Here $x_0 = 0.6$ and $h = 0.1$. There are only two values of q ($q = \pm 2$) for which the given data would suffice. With $q = 2$, (10.38) would assume the form

$$\begin{aligned} y'(0.6) &\approx \frac{1}{2^2(0.1)(2^2-1)} \left[2^3 y(0.7) - y(0.8) - 2^3 y(0.5) + y(0.4) \right] \\ &= \frac{1}{1.2} \left[8(2.3275054) - 2.6510818 - 8(1.7974426) + 1.5836494 \right] \\ &= 2.644225. \end{aligned}$$

Similarly (10.39) takes the form

$$\begin{aligned} y''(0.6) &\approx \frac{1}{(0.1)^2 2^2 (2^2 - 1)} \left[2^4 y(0.7) - y(0.8) - (2 \times 2^4 - 2) y(0.6) + 2^4 y(0.5) - y(0.4) \right] \\ &= \frac{1}{0.12} \left[16(2.3275054) - 2.6510818 - 30(2.0442376) + 16(1.7974426) - 1.5836494 \right] \\ &= 3.64424. \end{aligned}$$

EXERCISE 10

1. Use a numerical differentiation formula based on Newton's forward differences to estimate $y'(50)$ and $y''(50)$ from the data

x	50	55	60	65
y	1.6990	1.7404	1.7782	1.8129

Given that $y = \log_{10} x$, compare your result with the exact value, also find actual error.

2. The path $y = f(t)$ described by a point moving in a straight line is given by the following table:

t in sec	0.00	0.01	0.02	0.03	0.04	0.05
y in cm	0.00	1.519	6.031	13.397	23.396	35.721
t in sec	0.06	0.07	0.08	0.09		
y in cm	50.000	65.798	82.635	100.000		

Using forward differences upto order 5, approximate the velocity and acceleration of the point at $t = 0, 0.01$.

3. The points (x_0, y_0) , (x_1, y_1) , (x_2, y_2) and (x_3, y_3) , the x 's being equally spaced, lie on a cubic curve. Show that the approximate value of dy/dx at (x_1, y_1) is

$$-\frac{1}{6h}[y_3 - 6y_2 + 3y_1 + 2y_0].$$

4. Approximate $y'(65)$ and $y''(65)$ taking the data given in problem 1 and using Newton's backward differences.
5. Approximate velocity and acceleration of the point at $t = 0.08$ using backward differences upto order 5 and data of the problem 2.
6. Approximate $f'(x)$ and $f''(x)$ at $x=23.8$ if $f(x)$ is given by the following table:

x	5.0	10.0	15.0	20.0
f(x)	1.6094	2.3026	2.7081	2.9957

7. Estimate $y'(1)$ and $y''(1)$ for the function $y=y(x)$ given by the following table and using Stirling's formula.

x	0.96	0.98	1.00	1.02	1.04
y	0.7825361	0.7739332	0.7651977	0.7563321	0.7473390

8. Use Stirling's formula to approximate $f'(0.5)$ taking differences upto order 3, of a function tabulated below. Also, find the value of x at which $f(x)$ has its maximum.

x	0.3	0.4	0.5	0.6	0.7	0.8
$f(x)$	0.20819	0.23777	0.24958	0.24582	0.22920	0.20276

9. Approximate $f'(0.5)$ using Bessel's formula and the data given in problem 7.
10. A function $y = f(x)$ is specified by the following data:

x	0	0.2	0.4	0.6	0.8	1.0	1.2
y	0.710	1.175	1.811	2.666	3.801	5.292	7.232

Find radius of curvature at the point $x=0.6$ using Stirling's formula.

11. If a function $y=f(x)$ is given by the following data. Find the value of x for which y is minimum.

x	3	4	5	6	7	8
y	-205	-240	-259	-262	-250	-224

12. Find the location and value of $y_{\max}$ if a function $y = f(x)$ is given by the following data

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2
y	-1.651	0.300	2.178	3.505	3.800	2.582	-0.627

13. Calculate $f'(0.12)$ when $f(x)$ is given by the table

x	0.05	0.10	0.20	0.26
$f(x)$	0.0500	0.0999	0.1987	0.2571

14. Use Lagrange's formula to approximate $f'(-0.8)$ from the data

x	-1.0	-0.8	-0.6	-0.4
$f(x)$	0.3679	0.4493	0.5488	0.6703

Given that $f(x) = e^x$, obtain error bound on your result. Compare the result with the exact answer.

15. Derive the three-point formula $f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h}$

by Lagrange's formula. [Hint: Take interpolation points $-h, 0,$

h , and $x_0 = 0$ for convenience].

16. Approximate $f'(1.8)$ and $f''(1.8)$ correct to six decimal places, using three-point formulae (10.31) and (10.32) for $h=0.01$. Compare your results to the correct value if $f(x)=e^x$.
17. Let $f(x) = 2^x \sin x$. Compute approximations to $f'(1.05)$ and $f''(1.05)$ with $h = .05$ and $h = .01$ in the formulae (10.31) and (10.32) and using the following data:

x	1.0	1.04	1.06	1.10
$f(x)$	1.6829420	1.7732994	1.8188014	1.9103448

18. Derive the three-point formula

$$f'(x_0) = \frac{4f(x_0+h) - 3f(x_0) - f(x_0+2h)}{2h} + \frac{h^2}{3} f'''(\xi)$$

by Taylor's expansion.

[Hint: Use Taylor's expansion for $f(x_0+h)$ & $f(x_0+2h)$].